

## Academic Positions

### Columbia University

Founder's Postdoctoral Fellow, Department of Statistics

New York, NY

2025–present

## Education

### University of Pennsylvania

Ph.D. in Statistics, The Wharton School

Advisor: T. Tony Cai

Philadelphia, PA

2020–2025

### Indian Statistical Institute

M.Stat., specialization in Probability

**Ranked first**; First Division with Distinction

Kolkata, India

2018–2020

### Indian Statistical Institute

B.Stat. (Honours)

**Ranked first**; First Division with Distinction

Kolkata, India

2015–2018

## Publications

**Author notation:** ( $\alpha\beta$ ) Alphabetical ordering. \* Equal contribution.

1. ( $\alpha\beta$ ) T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. Optimal federated learning for functional mean estimation under heterogeneous privacy constraints. In press, *Journal of the Royal Statistical Society: Series B*, 2026. [paper]
2. ( $\alpha\beta$ ) T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. Optimal federated learning for nonparametric regression with heterogeneous distributed differential privacy constraints. In press, *Journal of the American Statistical Association*, 2026. [paper]
3. **Abhinav Chakraborty**, Sayak Chatterjee, and Sagnik Nandy. PriME: Privacy-aware membership profile estimation in networks. In press, *SIAM Journal on Mathematics of Data Science*, 2026. [paper]
4. ( $\alpha\beta$ ) T. Tony Cai, **Abhinav Chakraborty**, and Yichen Wang. Optimal differentially private ranking from pairwise comparisons. *Journal of the American Statistical Association*, 2026. [paper]
5. ( $\alpha\beta$ ) Arnab Auddy, T. Tony Cai, and **Abhinav Chakraborty**. Minimax and adaptive transfer learning for nonparametric classification under distributed differential privacy constraints. *Journal of the Royal Statistical Society: Series B*, 88(3):903–929, 2026. [paper]
6. **Abhinav Chakraborty**, Arnab Auddy, and T. Tony Cai. When data can't meet: Estimating correlation across privacy barriers. *Advances in Neural Information Processing Systems*, 38, 2025. (Spotlight.) [paper]
7. Ziang Niu\*, **Abhinav Chakraborty**\*, Oliver Dukes, and Eugene Katsevich. Reconciling model-X and doubly robust approaches to conditional independence testing. *The Annals of Statistics*, 52(3):895–921, 2024. [paper]
8. **Abhinav Chakraborty**\*, Anirban Chatterjee\*, and Abhinandan Dalal\*. PrIsing: Privacy-preserving peer effect estimation via Ising model. *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, PMLR 238:2692–2700, 2024. [paper]

## Preprints

1. Lucas Kania\*, **Abhinav Chakraborty**\*, Edward Kennedy, Larry Wasserman, and Sivaraman Balakrishnan. Optimal inference with black-box predictions. arXiv preprint arXiv:2608.10155, 2026. [paper]
2. **Abhinav Chakraborty** and Sagnik Nandy. Transfer learning in high-dimensional clustering: Minimax thresholds and applications in single-cell data. arXiv preprint arXiv:2607.25031, 2026. [paper]
3. **Abhinav Chakraborty** and Subha Maity. The statistical cost of adaptation in multi-source transfer learning. arXiv preprint arXiv:2605.09471, 2026. [paper]

4. **Abhinav Chakraborty\***, Yuetian Luo\*, and Rina Foygel Barber. Stability and accuracy trade-offs in statistical estimation. arXiv preprint arXiv:2601.11701, 2026. Under major revision at *IEEE Transactions on Information Theory*. [\[paper\]](#)
5. **Abhinav Chakraborty\***, Junu Lee\*, and Eugene Katsevich. Comparing three learn-then-test paradigms in a multivariate normal means problem. arXiv preprint arXiv:2601.07764, 2026. [\[paper\]](#)
6. ( $\alpha\beta$ ) T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. The cost of adaptation under differential privacy: Optimal adaptive federated density estimation. arXiv preprint arXiv:2512.14337, 2025. [\[paper\]](#)
7. Sayak Chatterjee, Anirban Chatterjee, **Abhinav Chakraborty**, and Bhaswar B. Bhattacharya. Asymptotic normality of subgraph counts in sparse inhomogeneous random graphs. arXiv preprint arXiv:2512.12937, 2025. [\[paper\]](#)
8. **Abhinav Chakraborty\***, Jeffrey Zhang\*, and Eugene Katsevich. Doubly robust and computationally efficient high-dimensional variable selection. arXiv preprint arXiv:2409.09512, 2024. [\[paper\]](#)
9. ( $\alpha\beta$ ) T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. Federated nonparametric hypothesis testing with differential privacy constraints: Optimal rates and adaptive tests. arXiv preprint arXiv:2406.06749, 2024. R&R at *JMLR*. [\[paper\]](#)
10. **Abhinav Chakraborty**, Soumendu Sundar Mukherjee, and Arijit Chakrabarti. High dimensional PCA: A new model selection criterion. arXiv preprint arXiv:2011.04470, 2020. [\[paper\]](#)

## Awards and Honors

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|-----------------------------------------------------------------------------------------------------------------------|------------|
| <b>IMS Travel Award</b><br>For the Frontiers in Statistical Machine Learning Workshop.                                | 2025       |
| <b>Travel Award, Optimization and Statistical Learning Workshop</b><br>Columbia University.                           | 2025       |
| <b>Lawrence D. Brown Best Student Paper Award</b><br>The Wharton School, University of Pennsylvania.                  | 2024       |
| <b>George James Term Award</b><br>Institute of Mathematical Statistics Annual Meeting.                                | 2022, 2024 |
| <b>ISI Alumni Association J. K. Ghosh Memorial Gold Medal</b><br>Top scorer in the M.Stat. program.                   | 2020       |
| <b>P. C. Mahalanobis Gold Medal</b><br>Outstanding M.Stat. student; best master's dissertation.                       | 2020       |
| <b>ISI Alumni Association Mrs. M. R. Iyer Memorial Gold Medal</b><br>Top scorer in the B.Stat. program.               | 2018       |
| <b>Nikhilesh Bhattacharya Gold Medal</b><br>Highest marks in Statistics in the B.Stat. (Hons.) program.               | 2018       |
| <b>INSPIRE Scholarship</b><br>Awarded to students in the top 1% of the Indian School Certificate Examination.         | 2015       |
| <b>Indian National Mathematics Olympiad qualifier</b><br>Top-30 candidate in the Regional Mathematics Olympiad, 2013. | 2014       |

## Talks

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|-----------------------------------------------------------------------------------------------------|------|
| <b>MCP Conference, Philadelphia, PA</b><br>Invited talk: Learn-then-test procedures.                | 2025 |
| <b>IISA Conference, Lincoln, NE</b><br>Invited talk: Federated transfer learning.                   | 2025 |
| <b>IMS Annual Meeting, Bochum, Germany</b><br>Federated nonparametric regression.                   | 2024 |
| <b>Ph.D. Student Seminar, University of Pennsylvania</b><br>Optimal distributed private estimators. | 2023 |
| <b>IMS Annual Meeting, London, UK</b><br>High dimensional PCA.                                      | 2022 |
| <b>P. C. Mahalanobis Gold Medal Presentation, Indian Statistical Institute</b>                      | 2020 |

Model selection in high dimensional PCA.

**D. Basu Memorial Presentation, Indian Statistical Institute**

2018

MCMC methods in Bayesian inference.

## Teaching and Mentoring

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**Teaching Assistant, The Wharton School, University of Pennsylvania**

2021–2024

Introduction to Probability (STAT 4300); Probability Theory (MATH 6480); Mathematical Statistics (STAT 971); Applied Econometrics (STAT 520).

**Ph.D. Peers Program, The Wharton School, University of Pennsylvania**

2023

Mentored an incoming Wharton Ph.D. student in navigating the program and transitioning to Penn.

## Professional Service

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**Journal reviewer:** *The Annals of Statistics*; *Journal of the American Statistical Association (JASA)*; *IEEE Transactions on Information Theory*; *Journal of the Royal Statistical Society: Series B (JRSSB)*; and *Biometrika*.